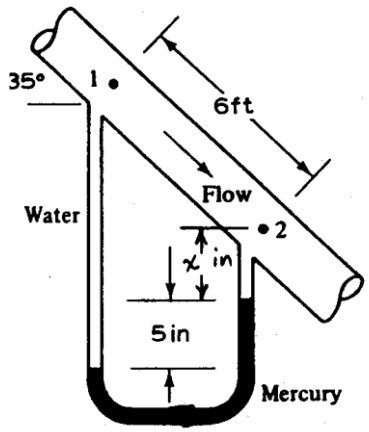
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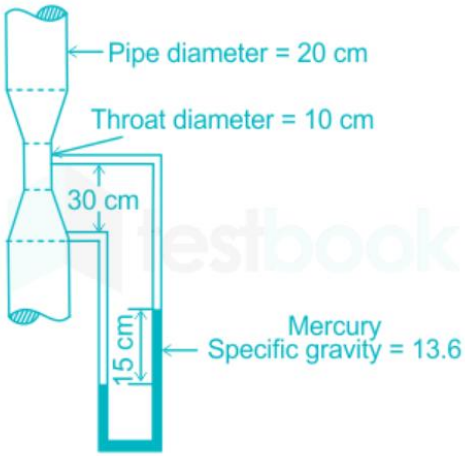
III SEMESTER B.TECH. (BIOTECHNOLOGY)
END-SEMESTER MAKEUP EXAMINATION, DEC/JAN 2025
SUBJECT: FLUID FLOW OPERATIONS (BIO 2124)

REVISED CREDIT SYSTEM

ANSWER ALL QUESTIONS

TIME: 3 HOURS **MAX. MARKS: 50**

Q. NO	QUESTIONS	M	CO	PO	BTL
1A	An open U tube manometer is being used to measure the absolute pressure p_a in a vessel containing air. The pressure p_b is atmospheric pressure which is 754 mmHg. The liquid in the manometer is water having a density of 1000kg/m^3. Assume that the density of air is 1.3 kg/m^3 and that the distance Z is very small. The reading R is 0.415 m. Calculate p_a in kPa.	**4**	**1**	**1-3**	**4**
1B	Water flows downward in a pipe at 35°, as shown in fig. The pressure drop P1 – P2 is partly due to gravity and partly due to friction. The mercury manometer reads a 5-in height difference. What is the total pressure drop P1 –P2? What is the pressure drop due to friction only between 1 and 2? Does the manometer reading correspond only to friction drop? 	**4**	**1**	**1-3**	**3**
1C	A differential manometer shows a higher column of liquid in one limb compared to the other. What does this indicate about the pressure difference between the two points being measured?	**2**	**1**	**1-3**	**4**
2A	The lower plate is being pulled at a relative velocity of 0.40 m/s greater than the top plate. The fluid used is water at 25 ^{o}C. a. How far apart should the two plates be placed so that the shear stress is 0.30 N/m^2. Also calculate the shear rate. b. If oil with a viscosity of 2.0 x 10^{-2} Pa.s is used instead at the same plate spacing and velocity as in part (a). What is the shear stress and the shear rate?	**4**	**1**	**1-3**	**3**

2B	A venturimeter as shown in the figure is connected to measure the flow of water in a verticle pipe of 20 cm diameter. Assume g = 9.8 m/s^2. When the deflection in the mercury manometer is 15 cm, calculate the flowrate considering no loss in the venturimeter?  	**4**	**2**	**1-3**	**4**
2C	In an open tank draining through a pipe, how does the height of the fluid above the pipe affect the flow velocity at the pipe outlet?	**2**	**2**	**1-3**	**3**
3A	An oil having a density of 833 kg/m^3 amd a viscosity od 3.3 x 10^{-3} Pa.s is pumped from an open tank to a pressurized tank held at 345 kPa gage. The oil is pumped from an inlet at the side of the open tank through a line of commercial steel pipe having an inside diameter of 0.07792 m at the rate of 3.494 x 10^{-3} m^3/s. the length of straight pipe is 122 m and the pipe contains two elbows (90°) and a globe valve half open. The level of the liquid in the open tank is 20 m above the liquid level in the pressurized tank. Calculate the total frictional loss in J/kg. K_f (globe valve, half open) = 9.5 K_f (elbow) = 0.75; K_c = 0.55; K_{ex} = 1	**4**	**3**	**1-3**	**4**
3B	Water at 20ºC is pumped from a storage tank through 100 m of 3 cm diameter pipe. The pipe line has TWO globe valves which are fully open and THREE 90º elbows. Water is discharged into another tank through a spray nozzle. The discharge is at a height of 20 m above the level of water in the storage tank. The pressure required at the nozzle entrance is 4x10^5 N/m^2. Flow rate of water 1kg/sec. Viscosity is 0.975 cP $f = 0.0014 + 0.125/Re^{0.32}$ Estimate Energy loss due to friction. Data: Equivalent length in terms of pipe diameter: Open globe valve = 300D; 90º Elbow =30D	**4**	**3**	**1-3**	**4**
3C	Consider two pipes with the same length and pressure difference. One pipe has twice the radius of the other. How much greater is the flow rate in the wider pipe? Assume flow is laminar.	**2**	**3**	**1-3**	**5**
4A	Calculate the pressure drop of air flowing at 30 ºC and 1 atm pressure through a bed of 1.25 cm diameter spheres, at a rate of 60 kg/min. The bed is 125 cm diameter and	**4**	**4**	**1-3**	**4**

	250 cm height. The porosity of the bed is 0.38. The viscosity of air is 0.0182 cP and the density are 0.001156 g/cc.				
4B	Calculate the sphericity of a solid particle of a cubical shape.	**2**	**4**	**1-3**	**4**
4C	A fermentation broth with viscosity 10^{-2} Pa-s and density 1000 kg/m^3 is agitated in a 50m^3 baffled tank using a marine propeller (refer curve 5) 1.3 m in dia. Calculate the power required for a stirred speed of 4 rps.	**4**	**4**	**1-3**	**5**
5A	Solid particles having a size of 0.12 mm, a shape factor of 0.88 and a density of 1000 kg/m^3 are to be fluidized using air at 2 atm abs and 25 ºC. The voidage at min. fluidization is 0.42. If the cross section of the empty bed is 0.3 m^2 and the bed contains 300 kg of solid, calculate: a. the min. height of the fluidized bed. b. the pressure drop at min. fluidization conditions. c. the min. velocity for fluidization. Data: Air properties at 2 atm abs and 25ºCare: Viscosity=1.845x10^{-5} Pa.s ; density = 2.374 kg/m^3.	**4**	**4**	**1-3**	**4**
5B	To clean a sand-bed filter, it is fluidized at minimum conditions using water at 24ºC. The round sand particles having a density of 2550 kg/m^3 and an average size of 0.40 mm. The sand has the sphericity of 0.86. the void fraction at minimum fluidizing conditions is 0.42. a. The bed diameter is 0.4 m and the desired height of the bed at these minimum fluidizing conditions is 1.75 m. Calculate the amount of solids needed. b. Calculate the pressure drop at these conditions and the minimum velocity for fluidization.	**4**	**4**	**1-3**	**4**
5C	Why does the pressure drop across a fluidized bed remain constant after achieving minimum fluidization?	**2**	**4**	**1-3**	**5**
	CO: Course Outcome; BLOOM TAXONOMY LEVEL: 1-Remember, 2-Understanding, 3-Application, 4-Analyzis, 5-Evaluation, 6-Creation				